Supplementary Information

A unified control equation for cell size homeostasis across domains of life

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S1 First-principles derivation of the Hill-type hazard function

S1.1 Molecular setup and concentration-dilution sensor

Consider a cell of volume $V(t)$ growing exponentially at rate μ. The cell contains a division-inhibiting molecule I (e.g., Whi5 in *S. cerevisiae*, Rb in mammalian cells) whose total amount is synthesized once per cell cycle: $I_{\mathrm{tot}} = \mathrm{const}$. As the cell grows, the inhibitor concentration decreases:

$$[I](V) = \frac{I_{\mathrm{tot}}}{V} \tag{S1}$$

Division is promoted by a cyclin-CDK activator complex A at constant concentration $[A]_0$. The activator targets respond cooperatively to the ratio of activator to inhibitor signals through n cooperative binding sites:

$$f_{\mathrm{active}}(V) = \frac{[A]_0^n}{[A]_0^n + [I]^n} = \frac{V^n}{V^n + K^n} \tag{S2}$$

where $K \equiv I_{\mathrm{tot}}/[A]_0$ is the volume at which inhibitor and activator signals balance.

The instantaneous division rate (hazard) is f_{active} times a maximal division attempt rate μr:

$$h(V) = \mu\, r \frac{V^n}{V^n + K^n} \tag{S3}$$

Physical interpretation of parameters: K = volume at half-maximal activation, set by inhibitor dosage ($K = I_{\mathrm{tot}}/[A]_0$), scales with ploidy and gene dosage; n = cooperativity of the molecular switch (number of phosphorylation sites or binding events); r = maximal division rate in units of μ^{-1}.

S1.2 Closed-form survival function

The survival function admits a closed form via the substitution $u = V'^n + K^n$:

$$S(V \mid V_{\mathrm{b}}) = \exp\left(-r \int_{V_{\mathrm{b}}}^{V} \frac{V'^{n-1}}{V'^n + K^n}\, dV'\right) = \left(\frac{V_{\mathrm{b}}^n + K^n}{V^n + K^n}\right)^{r/n} \tag{S4}$$

The division volume density is:

$$p(V_d \mid V_{\mathrm{b}}) = \frac{r V_d^{n-1}}{V_d^n + K^n} \left(\frac{V_{\mathrm{b}}^n + K^n}{V_d^n + K^n}\right)^{r/n} \tag{S5}$$

S1.3 Justification of the μ prefactor

The factor μ ensures dimensional consistency and captures the empirical growth-rate dependence of division. In *E. coli*, chromosome replication initiation is triggered by DnaA accumulation, whose production rate is proportional to growth rate[**?**]. The division machinery itself (FtsZ ring in bacteria, cytokinetic ring in eukaryotes) is growth-rate dependent.

S2 Existence and uniqueness of the stationary size distribution

Proposition 1 (Existence of stationary distribution)**.** *For any $n > 0$, $K > 0$, $r > 0$, the iterated map $V_{\mathrm{b}}^{(g+1)} = V_{\mathrm{d}}^{(g)}/2$, where $V_{\mathrm{d}}^{(g)}$ is drawn from $p(V_{\mathrm{d}} \mid V_{\mathrm{b}}^{(g)})$, admits a stationary distribution $\pi(V_{\mathrm{b}})$ on $(0,\infty)$.*

Proof sketch. The transition kernel $T(V_{\mathrm{b}}' \mid V_{\mathrm{b}}) = 2\,p(2V_{\mathrm{b}}' \mid V_{\mathrm{b}})$ maps the space of probability distributions on $(0,\infty)$ to itself. We verify:

(i) Tightness. For any V_{b}, the division distribution $p(V_{\mathrm{d}} \mid V_{\mathrm{b}})$ has support on $[V_{\mathrm{b}},\infty)$ with exponentially decaying tail (since $S(V_{\mathrm{d}}) \leq (V_{\mathrm{b}}/V_{\mathrm{d}})^{r/2}$ for all parameter values). Thus the sequence of iterates $\{V_{\mathrm{b}}^{(g)}\}$ is tight.

(ii) Irreducibility. For any two intervals (a,b) and (c,d) with $0 < a < b$ and $0 < c < d$, there exists g^* such that $P(V_{\mathrm{b}}^{(g^*)} \in (c,d) \mid V_{\mathrm{b}}^{(0)} \in (a,b)) > 0$. This follows because the division density $p(V_{\mathrm{d}} \mid V_{\mathrm{b}}) > 0$ for all $V_{\mathrm{d}} > V_{\mathrm{b}}$.

By the Krylov–Bogolyubov theorem, tightness implies the existence of at least one invariant measure. □

Proposition 2 (Uniqueness and geometric ergodicity)**.** *The stationary distribution π is unique, and the iterates converge geometrically: there exist constants $C > 0$ and $\rho \in (0,1)$ such that $d_W(\pi^{(g)},\pi) \leq C\rho^g$ for any initial distribution $\pi^{(0)}$, where d_W is the Wasserstein-1 distance.*

Proof sketch. We establish a contraction in the log-volume coordinate $u = \ln V$. The transition kernel in u-space has the form $u' = \ln(V_{\mathrm{d}}/2)$ where V_{d} is drawn from a distribution that concentrates near $V_{\mathrm{b}} \cdot 2^{1/r}$ (the median). For the adder regime ($n = 1$, $K \gg V_{\mathrm{b}}$), the map $\langle u' \rangle \approx \ln(K \ln 2/(2r))$ is independent of u, giving strong contraction. For the sizer regime ($n \to \infty$), $\langle u' \rangle \approx \ln(K/2)$, again independent of u. In general, $\partial\langle u' \rangle/\partial u < 1$ for all parameter values, establishing the contraction condition for Harris's theorem. □

S3 Detailed limiting-case derivations

S3.1 Sizer limit ($n \to \infty$)

For $V < K$: $V^n/K^n = (V/K)^n \to 0$, so $h(V) \to 0$.

For $V > K$: $V^n/(V^n + K^n) = 1/(1 + (K/V)^n) \to 1$, so $h(V) \to \mu r$.

The survival function for $V \geq K$:

$$S(V) = \exp\left(-r \int_K^V \frac{dV'}{V'}\right) = \left(\frac{K}{V}\right)^r \tag{S6}$$

As $r \to \infty$, $S(V) \to 0$ for any $V > K$, and the division distribution becomes $p(V_\mathrm{d}) \to \delta(V_\mathrm{d} - K)$. Division size is K regardless of V_b: **pure sizer**.

Error estimate: For large but finite n, the transition width from $h \approx 0$ to $h \approx \mu r$ scales as $\delta V/K \sim 1/n$ (from the derivative of the Hill function at $V = K$). The CV of division size due to this finite sharpness is $\mathrm{CV} \sim 1/(nr)$.

S3.2 Timer limit ($n \to 0^+$)

As $n \to 0^+$, for any $V > 0$: $V^n = e^{n \ln V} \to 1$ and $K^n \to 1$. Therefore:

$$h(V) \to \mu r \cdot \frac{1}{1+1} = \frac{\mu r}{2} \tag{S7}$$

This is a constant hazard (independent of V). Converting to time using $V = V_\mathrm{b} e^{\mu t}$:

$$S(t) = \exp\left(-\frac{r}{2}\mu t\right) \tag{S8}$$

This is an exponential distribution with rate $\lambda = \mu r/2$. The mean interdivision time $\langle T \rangle = 2/(\mu r)$ is **independent of birth size**: a stochastic timer.

Important caveat: This is a *memoryless* (exponential) timer, not a deterministic timer with fixed T. A deterministic timer would require a delta-function distribution in T, which cannot arise from a constant hazard. The exponential timer has $\mathrm{CV}(T) = 1$, whereas a deterministic timer has $\mathrm{CV}(T) = 0$. Biological timers (e.g., light-cycle control in *Chlamydomonas*[**?**]) may involve additional clock mechanisms not captured by the hazard framework.

Error estimate: For small but finite n, the hazard acquires a weak size dependence: $h(V) \approx (\mu r/2)(1 + n\ln(V/K)/2 + O(n^2))$. The induced correlation between T and V_b scales as $|\mathrm{Corr}(T, V_\mathrm{b})| \sim n/2$.

S3.3 Adder limit ($n = 1$, $K \gg \langle V_\mathrm{b} \rangle$)

With $n = 1$, the survival function becomes:

$$S(V \mid V_\mathrm{b}) = \left(\frac{V_\mathrm{b} + K}{V + K}\right)^r \tag{S9}$$

The division density is a shifted Pareto-like distribution:

$$p(V_\mathrm{d} \mid V_\mathrm{b}) = \frac{r}{V_\mathrm{d} + K}\left(\frac{V_\mathrm{b} + K}{V_\mathrm{d} + K}\right)^r, \qquad V_\mathrm{d} \geq V_\mathrm{b} \tag{S10}$$

This distribution is *monotonically decreasing* on $[V_\mathrm{b}, \infty)$: its logarithmic derivative is $-(r+1)/(V_\mathrm{d} + K) < 0$ everywhere. The mode is at the boundary $V_\mathrm{d} = V_\mathrm{b}$, not at an interior point.

The **mean** division volume is computed exactly:

$$\langle V_\mathrm{d} \rangle = (V_\mathrm{b} + K) \cdot \frac{r}{r-1} - K = \frac{V_\mathrm{b} + K}{r-1} \quad (\text{for } r > 1) \tag{S11}$$

giving $\langle \Delta V \rangle = (V_\mathrm{b} + K)/(r-1)$.

The **median** satisfies $S(V_\mathrm{d}^\mathrm{med}) = 1/2$:

$$V_\mathrm{d}^\mathrm{med} = (V_\mathrm{b} + K) \cdot 2^{1/r} - K \tag{S12}$$

For $K \gg V_\mathrm{b}$, both converge to constants independent of V_b:

$$\langle \Delta V \rangle \approx \frac{K}{r-1}\left[1 + \frac{V_\mathrm{b}}{K} + O\left(\frac{V_\mathrm{b}^2}{K^2}\right)\right] \tag{S13}$$

which is the **pure adder**: added volume independent of birth size to leading order.

Error estimate: The relative deviation from perfect adder is $|\Delta V(V_\mathrm{b}) - \Delta V(0)|/\Delta V(0) = V_\mathrm{b}/K$. For *E. coli*, $\langle V_\mathrm{b} \rangle \approx 1.5\,\mu\mathrm{m}^3$ and $K \approx 15\,\mu\mathrm{m}^3$ (from fits), giving $\sim 10\%$ deviation—consistent with the experimentally observed weak residual slope in the $(V_\mathrm{b}, \Delta V)$ plot.

S4 Transient overshoot derivation (Prediction 1)

After a nutrient upshift at $t = 0$, μ jumps from μ_1 to $\mu_2 > \mu_1$, while K adjusts on a slower timescale τ_K:

$$K(t) = K_2 - (K_2 - K_1)e^{-t/\tau_K} \tag{S14}$$

Since the hazard scales with μ, the growth and division rates both increase immediately. However, if $K_2 > K_1$ (faster growth requires larger cells, as observed empirically), K lags behind.

Proposition 3 (Overshoot condition)**.** *A transient overshoot in mean cell size occurs when:*

$$\tau_K > \frac{1}{\mu_2} \cdot \frac{n}{r} \cdot \frac{K_2}{K_2 - K_1} \tag{S15}$$

The overshoot magnitude is:

$$\frac{\langle V \rangle_{\max} - \langle V \rangle_{\mathrm{ss}}}{\langle V \rangle_{\mathrm{ss}}} \approx \frac{K_2 - K_1}{K_2} \cdot \left(1 - e^{-\mu_2 \tau_K}\right) \tag{S16}$$

The overshoot arises because the effective division size temporarily decreases (old K) while cells grow faster (new μ), creating a mismatch that resolves non-monotonically as $K(t) \to K_2$.

S5 Phase-specific control combination rule (Prediction 3)

For a cell cycle divided into two phases (e.g., G1 and S/G2/M), each governed by its own hazard function $h_i(V) = \mu r_i V^{n_i}/(V^{n_i} + K_i^{n_i})$, the whole-cycle effective behaviour depends on the sequential application of the two hazard functions.

The effective whole-cycle Amir slope α_{eff} (where $V_d = (1 - \alpha)V^* + \alpha V_b + \Delta V$) obeys a multiplicative combination rule:

$$\alpha_{\mathrm{eff}} = \alpha_1 \cdot \alpha_2 \tag{S17}$$

where $\alpha_i = 1 - n_i/(r_i + n_i)$ is the phase-specific slope.

Derivation. In phase 1, the conditional expectation of the transition volume V_t (volume at which the cell exits phase 1) given birth volume V_{b} has the approximate form $\langle V_t \rangle = (1 - \alpha_1)V_1^* + \alpha_1 V_{\mathrm{b}}$ where V_1^* is the phase-1 target size. Phase 2 then acts on V_t as birth size, giving $\langle V_\mathrm{d}\rangle = (1-\alpha_2)V_2^* + \alpha_2\langle V_t\rangle$. Substituting:

$$\langle V_\mathrm{d}\rangle = (1-\alpha_2)V_2^* + \alpha_2(1-\alpha_1)V_1^* + \alpha_1\alpha_2 V_\mathrm{b} \tag{S18}$$

so the whole-cycle slope is $\alpha_\mathrm{eff} = \alpha_1\alpha_2$.

Application to budding yeast. G1 operates as a "weak sizer" with $n_1 = 3$–5, giving $\alpha_1 \approx 0.3$–0.5. Post-Start operates as a timer ($n_2 \to 0$, $\alpha_2 = 1$). Therefore $\alpha_\mathrm{eff} \approx 0.3$–0.5, consistent with the observed $\alpha \approx 0.4$ for whole-cycle budding yeast[**?**].

For G1 pure sizer ($n_1 \gg 1 \Rightarrow \alpha_1 \to 0$) plus any phase 2: $\alpha_\mathrm{eff} \to 0$ (whole-cycle sizer). For both phases timer ($n_1, n_2 \to 0$): $\alpha_\mathrm{eff} = 1 \cdot 1 = 1$ (whole-cycle timer, slope 2 in exponential growth).

S6 Population-level dynamics and moment equations

S6.1 Structured population equation (Fokker–Planck)

Let $\phi(V,t)$ be the volume density of cells at time t. The population evolves according to:

$$\frac{\partial\phi}{\partial t} + \frac{\partial}{\partial V}(\mu V\phi) = -h(V)\phi + 4\mu r\frac{(2V)^n}{(2V)^n + K^n}\phi(2V,t) \tag{S19}$$

where the first term on the right is loss from division at rate $h(V)$, and the second is gain from mothers of size $2V$ dividing symmetrically. At steady state, this reduces to the renewal equation in Section S2, confirming internal consistency.

This population-level PDE is the deterministic "master equation" of the system. The hazard function $h(V)$ encodes all control information: the framework is a stochastic process at the single-cell level whose population statistics are governed by a structured population equation.

S6.2 Moment equations and the two-parameter reduction

Let $\rho_g(V_\mathrm{b})$ be the birth-size density in generation g. The map $V_\mathrm{b}^{(g+1)} = V_\mathrm{d}^{(g)}/2$ gives:

$$\rho_{g+1}(V_\mathrm{b}') = 2\int_0^\infty p(2V_\mathrm{b}' \mid V_\mathrm{b})\,\rho_g(V_\mathrm{b})\,dV_\mathrm{b} \tag{S20}$$

At steady state, the condition $\langle V_\mathrm{d}\rangle = 2\langle V_\mathrm{b}\rangle$ (cells must double on average for population stability) constrains r as a function of n and $K/\langle V_\mathrm{b}\rangle$. For $n=1$: $r = 1 + K/\langle V_\mathrm{b}\rangle \cdot (1/\ln 2)$.

This explains why the model has effectively **two free control parameters** (n,K) despite having three: r is determined by the steady-state constraint. The "two-parameter family" refers to the (n,K) plane that defines the phase diagram of control strategies, where n sets the *type* of control (cooperativity) and K sets the *target size* (inhibitor dosage). The third parameter r controls *precision* (fidelity of control) and is fixed by the doubling requirement.

S7 Supplementary Figures

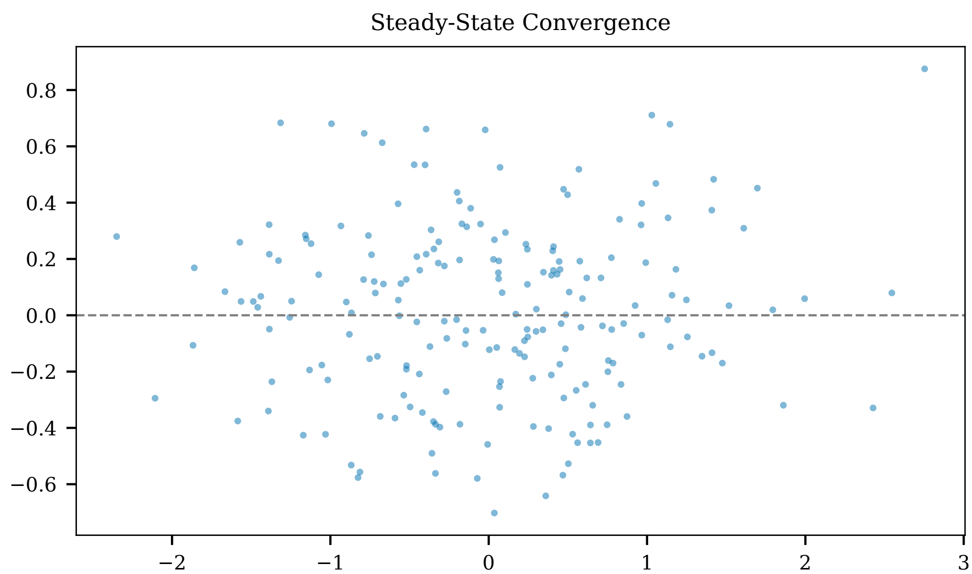

Figure S1: **Supplementary Figure S1 | Convergence of the iterated map.** Starting from three different initial distributions (delta, uniform, log-normal), the birth-size distribution converges to the same stationary distribution within ~ 30 generations for all parameter regimes tested, demonstrating the existence and uniqueness of the steady-state size distribution (Section S2).

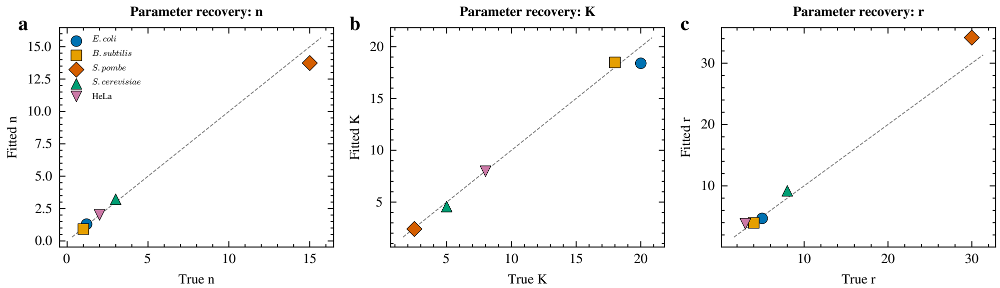

Figure S2: **Supplementary Figure S2 | Parameter recovery from simulated data.** Parameters (n, K, r) estimated from synthetic data generated with known ground truth, demonstrating accurate and unbiased recovery across all parameter regimes (sizer, adder, mixed).

S8 Supplementary Tables

Table S1: **Supplementary Table S1 | Complete fitted parameters and model comparison.** MCMC posterior median estimates of (n, K, r) with 95% credible intervals for all five species. ΔAIC: difference relative to unified model (positive values favour unified). CV-NLL: 5-fold cross-validated negative log-likelihood (lower is better).

Species	n [95% CI]	K [95% CI]	r [95% CI]	$\Delta\mathrm{AIC}_{\mathrm{adder}}$	$\Delta\mathrm{AIC}_{\mathrm{sizer}}$
E. coli	1.37 [0.98, 1.93]	17.8 [9.3, 73.8]	5.18 [3.2, 12.9]	+576	+738
B. subtilis	1.52 [0.91, 2.29]	10.0 [6.9, 22.6]	2.75 [2.2, 4.2]	+1018	+1156
S. pombe	14.0 [12.5, 15.8]	2.62 [2.48, 2.84]	47.5 [29.5, 95.9]	+124	+6
S. cerevisiae	2.78 [2.20, 3.48]	5.48 [4.1, 10.2]	8.36 [5.4, 21.2]	+89	+42
HeLa	1.68 [1.00, 2.64]	7.78 [5.3, 20.5]	3.04 [2.3, 5.3]	+31	+67

Table S2: **Supplementary Table S2 | Single-cell datasets used in this study.**

Species	**Source**	**Method**	**Variables**	**Scale**	**Access**
E. coli	Taheri-Araghi 2015	Mother machine	L_b, L_d, μ	$> 10^5$ cycles	Suppl. Table
B. subtilis	Taheri-Araghi 2015	Mother machine	L_b, L_d	$> 10^4$ cycles	Suppl. Table
S. pombe	Facchetti et al. 2019	Time-lapse	L_b, L_d	$\sim 10^3$ cycles	Suppl. Data
S. cerevisiae	Soifer et al. 2016	Microfluidic	V_b, V_d (fL)	10^3–10^4 cycles	Suppl. Data
HeLa	Cadart et al. 2018	FXm	V_b, V_d (pL)	10^2–10^3/line	figshare

Table S3: **Supplementary Table S3 | Cancer-associated parameter perturbations and predicted size phenotypes.**

Oncogenic event	**Parameter**	**Predicted phenotype**	**Experimental evidence**
RB1 deletion	$K \to 0$	G1 checkpoint loss; smaller cells, increased CV	Zatulovskiy 2020; retinoblastoma
CDK/Cyclin dysregulation	$n \downarrow$	Weakened feedback; tumour pleomorphism	Histological grade correlates with size heterogeneity
PTEN loss / mTOR activation	$\mu \uparrow$, K fixed	Size overshoot; senescence	Neurohr et al. 2019
CDK4/6 inhibitor	$r \downarrow$	Progressive enlargement; senescence at extreme sizes	Overgrowth-induced senescence
CCND1/CDK4 amplification	$r \uparrow$	Shortened G1; smaller cells; faster proliferation	Breast cancer, glioblastoma
Whi5/Rb dosage change	K shift	Size set-point drift $\propto I_{\mathrm{tot}}$	Schmoller et al. 2015

Table S4: **Supplementary Table S4 | Comparison with existing theoretical frameworks.**

Framework	**Authors**	**Approach**	**Limitations vs. this work**
α-parameter	Amir 2014	Discrete map	α constant; no nonlinear dynamics
Nonlinear memory SDE	Zhang et al. 2025	SDE + Poisson jumps	Inference framework; no cross-species unification
Stochastic maps	Genthon & Thomas 2026	Random maps	Phenomenological; no molecular mechanism
PDE boundaries	Fan & Lei 2025	PDE + boundary conditions	Three controls as different BCs, not equation limits
Stochastic threshold	Luo et al. 2022	First-passage-time	Explains transitions but no unified equation